H2 ANDERSON JUNIOR COLLEGE
HIGHER 2

CANDIDATE NAME

PDG

PDG INDEX NUMBER

I

BIOLOGY 9744/04

Paper 4 Practical

21 August 2018
Tuesday

Candidates answer on the Question Paper.
Additional Materials: As listed in the Confidential Instructions.

2 hours 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	
2	
3	
Total	**55**

This document consists of **18** printed pages.

Answer **all** the questions.

1 You are provided with a solution, labelled **E**, containing an enzyme which coagulates (clots) milk. Enzyme **E** hydrolyses (breaks) peptide bonds between certain amino acids in a protein found in milk and this result in the coagulation of the milk. Calcium ions are required for this coagulation.

When a mixture of milk, calcium chloride solution and **E** is gently rotated in a boiling tube, the coagulation goes through the stages shown in Fig. 1.1.

Stage 3 is the end-point of the enzyme-catalysed coagulation.

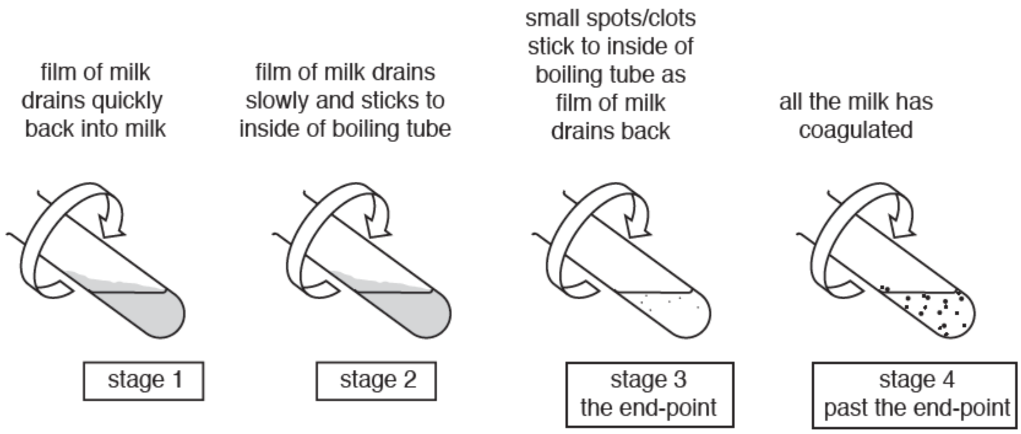

Fig. 1.1

You are provided with:

Labelled	Contents	Hazard	Volume/cm^3
C	20.0% calcium chloride solution	harmful irritant	50
W	distilled water	None	100
M	milk	None	100
E	1.0% enzyme solution	harmful irritant	10

If **C** or **E** comes into contact with your skin, wash off immediately under cold water. It is recommended that you wear suitable eye protection.

You are required to carry out a trial test (step **1** to step **9**) before you start your investigation.

*Read step **1** to step **9** before proceeding.*

Proceed as follows.

1 You are provided with a beaker labelled **water-bath**. Use the hot and cold water to set up a water-bath in this beaker. The starting temperature of the water-bath should be between 35°C and 40°C. You will **not** need to maintain this temperature during steps **2** to **9**.

2 Put $10 cm^3$ of **M** into a boiling tube, followed by $1 cm^3$ of **C**. Gently shake the boiling tube to mix **M** and **C**.

3 Put the boiling tube into the water-bath and leave for at least 3 minutes.

(a) **(i)** Explain why the boiling tube is left in the water-bath for at least 3 minutes in step **3**.

....

.... [1]

4 Remove the boiling tube from the water-bath.

5 The process of coagulation will start when **E** is added to the boiling tube. Using a syringe, put $1 cm^3$ of **E** into the boiling tube by gently pushing the plunger of the syringe so that **E** runs down the side of the boiling tube and forms a layer on the surface of the mixture, as shown in Fig. 1.2.

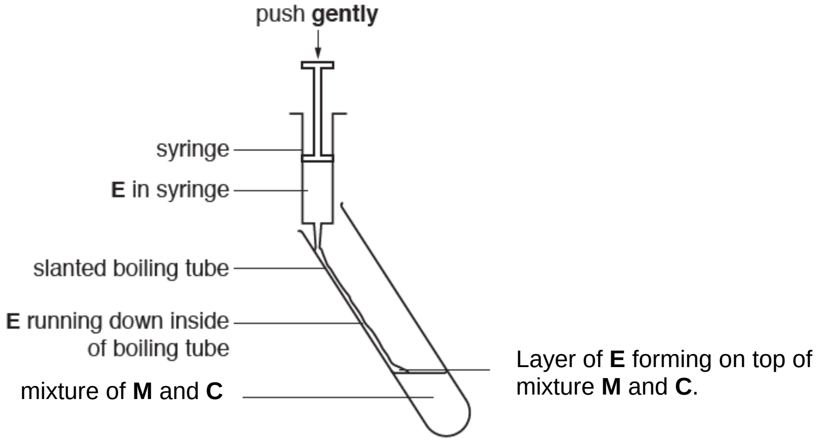

Fig. 1.2

6 Gently shake the boiling tube to mix the solutions and start timing.

7 Hold the boiling tube over a piece of black card on the table as shown in Fig. 1.3. Gently rotate the boiling tube to form a film of milk on the inside of the boiling tube.

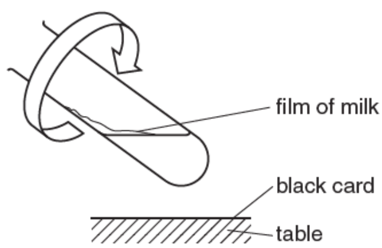

Fig. 1.3

8 Observe the film until the end-point is reached (**stage 3** in Fig. 1.1). Ignore any small bubbles on the inside of the boiling tube.

9 Record in **(a) (ii)** the time taken to reach the end-point.

If the end-point has not been reached in 4 minutes, stop the experiment and record 'more than 240'.

(ii)

Time taken to reach end-point.. [1]

(iii) A significant source of error for this investigation is deciding when the end-point is reached.

Suggest **one** advantage of carrying out this trial test.

…………………………………………………………………………………………………

………………………………………………………………………………………………… [1]

You are now required to investigate the effect of calcium chloride concentration of this enzyme-catalysed coagulation.

You will carry out a serial dilution of the 20.0% calcium chloride solution, **C**, to reduce the concentration of the calcium chloride solution by **half** between each of four successive dilutions, to give **C1**, **C2**, **C3** and **C4**. You will also need to set up a control, **T**.

You are required to make up at least 10 cm^3 of each concentration of calcium chloride solution in the small beakers or containers provided.

(b) **(i)** Complete Table 1.1 to show how you will make the concentrations of the calcium chloride solutions, **C1**, **C2**, **C3** and **C4**, and how you will set up the control, **T**.

Table 1.1

	C	**C1**	**C2**	**C3**	**C4**
Concentration of calcium chloride /%	20.0				
Label of calcium chloride to be diluted		C			
Volume of the calcium chloride solution to be diluted / cm^3					
Volume of the distilled water, **W**, to make up the dilution/ cm^3					
Description of the control, **T**: ………………………………………………………………………………………………… …………………………………………………………………………………………………					

[4]

10 Prepare the concentrations of calcium chloride as decided in **(b) (i).**

11 Adjust the temperature of the water-bath so that it is between 35°C and 40°C. You will need to maintain the water-bath at this temperature throughout the investigation.

12 Label six boiling tubes **C**, **C1**, **C2**, **C3**, **C4** and **T**, as in Table 1.1

13 Put 1 cm^3 of **C** into the appropriately labelled boiling tube and 10 cm^3 of **M** into the same boiling tube. Gently shake the boiling tube to mix **C** and **M**. Place the boiling tube in the water-bath.

14 Repeat step **13** using **C1**, **C2**, **C3** and **C4**.

15 Set up the control, **T**, and place it in the water-bath. Leave all the boiling tubes for at least 3 minutes.

*While you are waiting, read step **5** to step **9**.*

16 After 3 minutes remove one of the boiling tubes from the water-bath. Add 1cm^3 of **E** as in step **5**, then repeat step **6** to step **9** and record in **(b) (ii)** the time taken to reach the end-point.

17 Repeat step **16** with each of the other boiling tubes.

(ii) Record your results in an appropriate manner in the space below.

[4]

(iii) Calculate the rate of coagulation of milk when the concentration of calcium chloride was 20.0%.

Rate of coagulation………………………………. [1]

(iv) Suggest an explanation of the effect of calcium ions on the coagulation of milk by enzyme **E**.

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…………………………………………………………………………………………………… [2]

(v) Explain why temperature is a significant source of error in this investigation and suggest a modification to reduce or eliminate this error.

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(c) **(i)** This procedure can be modified to investigate other independent variables.

Describe how you could modify this procedure to investigate the effect of pH on the time taken to reach the end-point.

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…………………………………………………………………………………………………… [2]

(ii) Fig. 1.4 shows the results of the investigation done by a student.

pH 2 215 s	pH 3.5 15 s	pH 4.9 75 s
pH 2.7 60 s	pH 6.2 220 s	

Fig. 1.4

Plot, on the grid, a graph of the student's results in Fig. 1.4 to show the effect of pH on the time taken for milk to coagulate. Draw a line of best fit.

[3]

(iii) Estimate the time taken to reach the end-point at pH 5.3.

.... [1]

[Total: 22]

2 **N1** is a slide of a stained transverse section through a plant leaf.

You are not expected to be familiar with this specimen

You are required to:

- use the eyepiece graticule to measure depths of different tissues across the leaf
- use these measurements to find the simplest ratio of the depth of the leaf to the depth of the palisade layer, **P**
- draw a plan diagram of part of the leaf

(a) Select a part of the leaf on **N1** which shows the four layers **L** (**L1** and **L2**), **P** and **Q**.

Do **not** include a vascular bundle.

(i) Use the eyepiece graticule in the microscope to measure:
- the depth of the whole leaf, **T**
- the depth of each of the tissues, **L** (**L1** and **L2**), **P** and **Q**, as shown in Fig. 2.1.

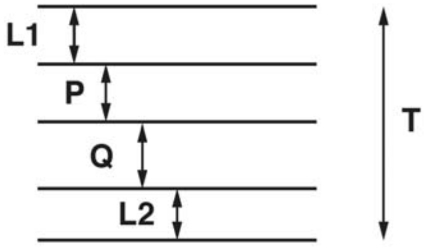

Fig. 2.1 (not drawn to scale)

T = ... eyepiece graticule units

L1 = ... eyepiece graticule units

P = ... eyepiece graticule units

Q = ... eyepiece graticule units

L2 = ... eyepiece graticule units [3]

(ii) Use the measurements from **(a) (i)** to determine the simplest ratio of the depth of the leaf (**T**) to the depth of the palisade layer (**P**).

You may lose marks if you do not show your working.

simplest ratio .... [3]

Use a sharp pencil for drawing.

(iii) Use the measurements from **(a) (i)** to help you draw a large plan diagram of a part of the leaf on **N1**, as shown by the shaded area in Fig. 2.2.

This must include at least **one** vascular bundle and **one** stoma.

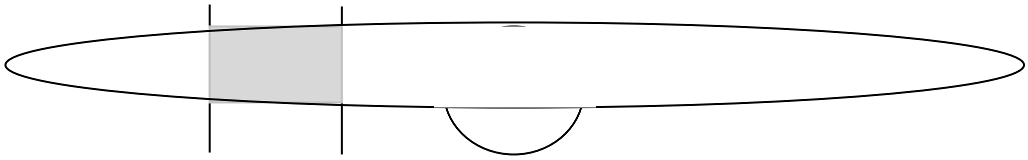

Fig. 2.2

You are expected to draw the correct shape and proportions of the different tissues.

Use **one** ruled label line and label **P** to identify the palisade layer.

[5]

(iv) Observe the upper epidermis of the leaf on **N1**. The upper epidermis has no guard cells.

Select one group of **four** adjacent (touching) cells from the upper epidermis.
Each cell must touch at least one of the other cells.

Make a large drawing of this group of **four** cells.

Use **one** ruled label line and the label **C** to identify a structure made of cellulose.

[4]

(b) Fig. 2.3 is a photomicrograph of a stained transverse section through part of a plant leaf from a different type of plant.

You are not expected to be familiar with this specimen.

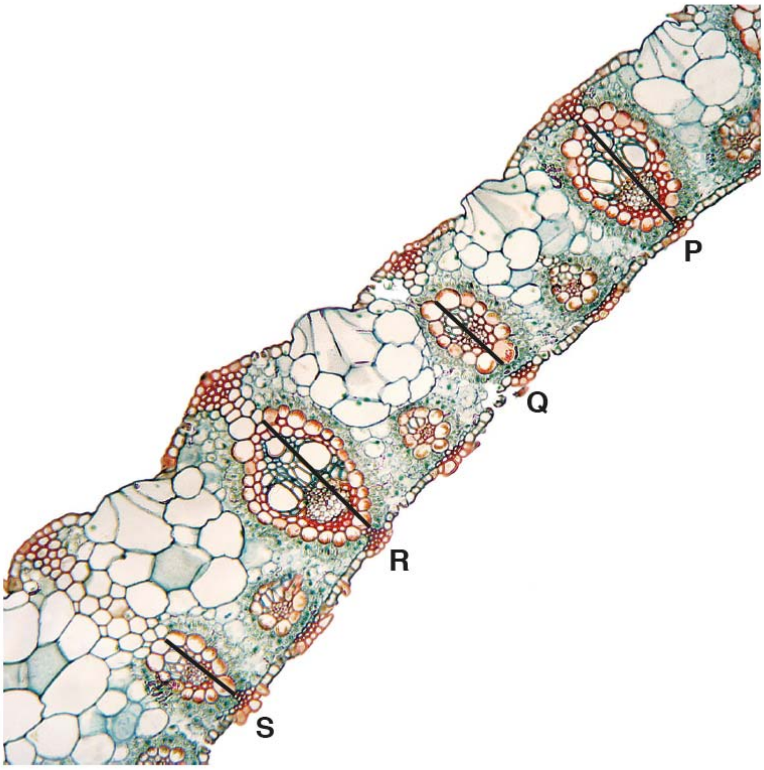

Fig. 2.3

(i) In Fig. 2.3 the lines **P**, **Q**, **R** and **S** are drawn across the length of four vascular bundles.

Describe how you would find out the **mean** actual length of these four vascular bundles.

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.... [2]

(ii) Observe the leaf on **N1** and the leaf in Fig. 2.3.

Use a suitable table to record observable differences between the specimen in Fig. 2.3 and the specimen on slide **N1**.

[2]

[Total: 19]

3 Planning question

Due to climate change, sea surface temperatures are expected to rise by about 2ºC in the worst case scenario. The current range of sea surface temperature locally is about 28ºC to about 30ºC. Changes in sea surface temperatures are expected to affect photosynthesis in aquatic plants.

The method used to find out the rate of photosynthesis is to measure the time taken for percentage concentration of carbon dioxide in solution to decrease below 0.04, causing hydrogencarbonate indicator solution to change colour.

You are provided with:

- solutions of three different percentage concentration of carbon dioxide in test tubes labelled **X**, **Y** and **Z**. **X** contains solution of more than 0.04% carbon dioxide. **Y** contains solution of 0.04% carbon dioxide. **Z** contains solution of less than 0.04% carbon dioxide.
- hydrogencarbonate indicator solution in a small beaker labelled **H**.

You are required to carry out tests on all three solutions to determine the colour change of the indicator in solutions with different percentage concentration of carbon dioxide.

Proceed as follows.

1. Put three drops of indicator solution into the test tubes **X**, **Y** and **Z**.

2. Swirl each tube.

(a) Record your results in Table 3.1

Tube	percentage concentration of carbon dioxide	colour
X	increasing above 0.04	
Y	0.04 (normal atmospheric)	
Z	falling below 0.04	

Table 3.1 [1]

Light can be provided by a bench lamp (which emits very little heat) placed at a distance away from the setup containing pondweed. It is observed that when the bench lamp is placed at a distance greater than 20 cm from the setup, the indicator will turn orange instead.

(b) Suggest why the indicator will turn orange when the bench lamp is placed at a distance greater than 20 cm from the setup.

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……… [1]

Using the information given and your own knowledge, design an experiment to investigate the effect of temperature on the rate of photosynthesis of pondweed (*Cabomba sp.*)

Your planning must be based on the assumption that you have been provided with the following equipment and materials, which you must use:

- Several pieces of pondweed (*Cabomba sp.*).
- Hydrogencarbonate indicator solution.
- Bench lamp which emit very little heat.
- Plastic tubes with screw-on-lids, which light can be pass through. The plastic tubes can contain 10 cm^3 of liquid in addition to the pondweed plant.
- Nutrient solution containing 0.04% carbon dioxide to immerse the pondweed in during the experiment
- Metre ruler
- Stopwatch
- Thermometer
- Boiling water
- A variety of different sized beakers, measuring cylinders or syringes for measuring volumes.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagram(s), if necessary
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- include description of a statistical method to determine if temperature change has indeed a significant impact on the rate of photosynthesis
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[12]

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[Turn over

[Turn over

THE END